

LLMDistill4Ads: Using Cross-Encoders to Distill from LLM Signals for Advertiser Keyphrase Recommendations at eBay

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Abstract

Sellers at eBay are recommended keyphrases to bid on to enhance the performance of their advertising campaigns. The relevance of these keyphrases is crucial in avoiding the overcrowding of search systems with irrelevant items and maintaining a positive seller perception. It is essential that keyphrase recommendations align with both seller and Search judgments regarding auctions. Due to the difficulty in procuring negative human judgment at scale, employing LLM-as-a-judge to mimic seller judgment has been established as the norm in several studies. This study introduces a novel two-step LLM distillation process from a LLM-judge used to debias our Embedding Based Retrieval (EBR) model from the various biases that exist in click-data. We distill from an LLM teacher via a cross-encoder assistant into a bi-encoder student using a multi-task training approach, ultimately employing the student bi-encoder to retrieve relevant advertiser keyphrases. We show that integrating a knowledge distillation process from LLMs in a multi-task training setup enhances bi-encoder performance in retrieving relevant advertiser keyphrases at eBay.

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1 Introduction

Within the domain of e-commerce, sellers strategically leverage online advertising methodologies, such as keyphrase recommendations [1, 34–36, 62], to mitigate their typically inferior positions within organic search outcomes. This tactic allows them to establish a strategic presence on the search results page (SRP) and enhance interaction with prospective buyers, see Figure 1. The relevance

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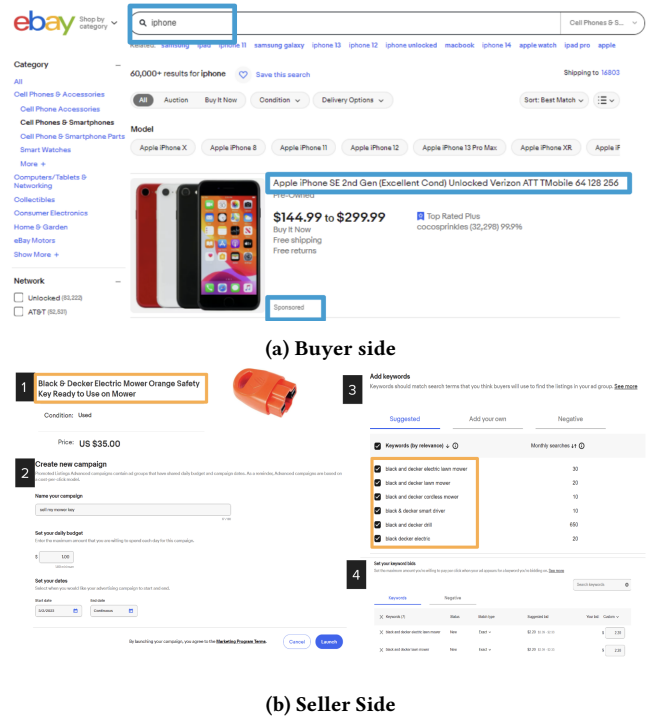


Figure 1: Screenshot of our keyphrases for manual targeting in Promoted Listings Priority for eBay Advertising.

of advertiser keyphrases is critical, as it influences seller perspectives and assists in avoiding the saturation of search systems with numerous non-relevant items competing for prominence in auctions. Models that assess advertiser keyphrase relevance are trained to identify general patterns in click and sales data. A keyphrase generating a substantial number of clicks or sales for an item suggests its relevance to the item. Essentially, clicks and sales serve as strong positive relevance indicators; however, they are inadequate for indicating non-relevance. E-commerce datasets suffer from missing-not-at-random (MNAR) conditions, due to a variety of biases [4, 7, 20, 21, 27, 46, 53, 60]. The lack of clicks for an item relative to a specific query does not automatically imply irrelevance. In the e-commerce context, buyers act as annotators, yet, unlike conventional annotators, they encounter a biased item presentation

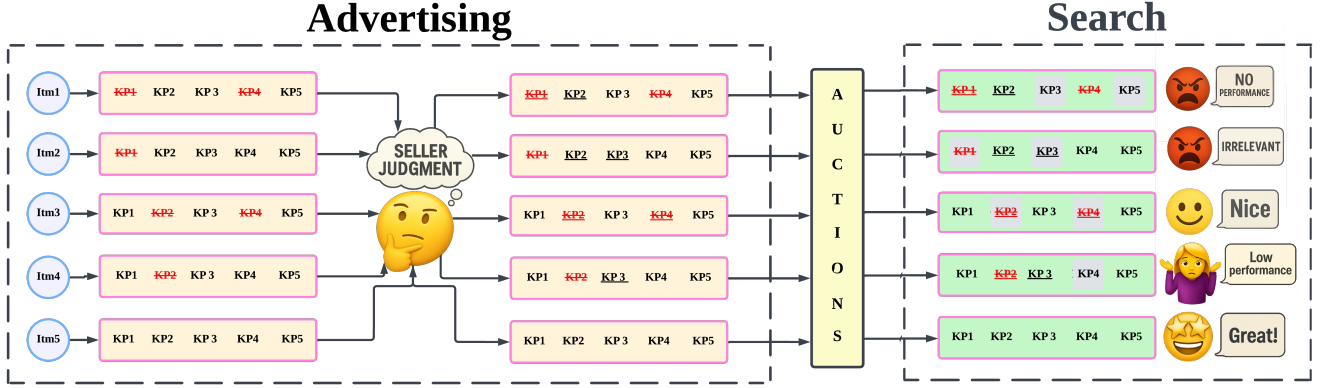


Figure 2: Auction mechanism of items (Itm) in relation to keyphrases (KP). Red strikethrough font represents filter of Advertising, the underline represents seller curation of keyphrases after advertising has filtered them while gray highlight represents the relevance filter of Search.

due to search rankings, which influences their selection in terms of clicks or sales. Consequently, an item with lower popularity will appear in a less favorable position on the SRP because search rankings are based on clicks, and consequently it may not receive any clicks or sales. This bias depreciates the dependability of negative relevance cues based solely on clicks or sales.

From the perspective of sellers, *eBay Advertising* promotes keyphrases to which they can bid on for their items. The items are then added to an auction matched to the keyphrase (query on the Search side). This auction process involves intricate interactions with *eBay Search*, which acts as an intermediary, aligning items with the advertised keyphrases proposed by *eBay Advertising* and applying a relevance filter to the auction items. Consequently, the logged click data encompasses only those keyphrases that pass the relevance filter (auction-winning, impression-gaining keyphrases are documented). *As a result, this introduces an additional bias, as training data contains only keyphrases approved by the Search relevance filter.* Training with this click data means the model is not exposed to keyphrases deemed irrelevant by Search, although Advertising does produce such keyphrases that need screening. This *middleman bias* constitutes a type of sample selection bias [12, 53], which further affects the click data within the domain of advertiser keyphrase recommendation.

The dynamics among Sellers, Advertising, and Search reflect an inherent asymmetry: Sellers have ultimate authority over adopting Advertising’s keyphrases, whereas Search possesses the final discretion to dismiss these keyphrases during auctions. For example, consider *Itm1* from Figure 2; while *eBay Advertising* finds *KP1* and *KP4* to be irrelevant, *eBay Search* deems *KP2*, *KP3*, and *KP5* irrelevant. Furthermore, the seller dismisses *KP2* as a valid keyphrase, consequently preventing *Itm1* from participating in any auctions. *This scenario unfolds irrespective of the keyphrases’ actual relevance; thus, even if *KP2*, *KP3*, and *KP5* are truly relevant, *Itm1* remains excluded from those auctions due to the seller and *eBay Search* deeming them irrelevant.* Perfect concordance is exemplified by *Itm5* and, to a lesser extent, *Itm3*, where all parties achieve consistent agreement. In the case of *Itm3*, the situation is not as favorable as for *Itm5*

since some keyphrases still do not enter auctions. Hence, an efficient retrieval model would not retrieve keyphrases *KP2* and *KP4* for *Itm3*, aligning with seller judgment, advertising judgment, and search judgment.

Understanding auction dynamics and mechanisms is crucial for effective advertising campaigns. Our product, Promoted Listings Priority, allows manual targeting where sellers can choose keyphrases or opt for our suggestions. For example, if sellers deem *KP2* for *Itm1* irrelevant, they can ignore it, nullifying our advice. This often leads to “irrelevant” keyphrases overwhelming sellers, lowering Seller Satisfaction and wasting resources for eBay. Offering seemingly sensible but ineffective keyphrases can reduce campaign efficiency and cause sellers to exit. Thus, it is essential that keyphrases align with human judgment and maintain strong performance. The relevance of Advertiser keyphrases can be viewed as a complex interaction among three systems: sellers’ judgment affecting adoption, Advertising providing bids and keyphrases, and Search managing keyphrase auctions as documented in previous work [10]. On the side of retrieval which is an upstream task, for efficiency purposes, the retrieved keyphrases must maximize all three relevance judgments while also driving revenue, sales and campaign performance for our sellers.

2 Related Work

As is common in the e-commerce domain, semantic search applications are often broken down into a three-stage process: candidates generation, relevance filtering and ranking. Recommending keyphrases for sellers to place bids on also follows this pattern. The candidates generation step consists in combining several recall models.¹ After this union of recalls, a relevance filtering step [11] is performed, followed by a ranker.

The advertiser keyphrase recommendations at eBay are generated using a zoo of models which include *fast-Text* (which creates word embeddings using the CBOW model and employs a straightforward linear neural network model with hierarchical softmax to improve the efficiency of training and inference processes), *Graphite*

¹The terms *recall model* and *candidates generator model* are used interchangeably.

[34] (which uses bipartite graphs to map words/tokens to the data points and then map them to the labels associated with the data points), *SL-emb* (which uses embeddings of the item's title to find similar listings and then recommends the related queries), or *Rules-based heuristic models* (which use simple techniques of storing item-keyphrase associations based on their co-occurrences in the search logs). Within this variety of recall solutions, Embedding-Based Retrieval (EBR) is a two-step solution which first embeds the buyers keyphrases and the item titles in a common vector space, and then recommends the k keyphrases closest to each item title in the vector space via an *Approximate Nearest Neighbor* search.

The two main architectures for encoders within the semantic search domain are cross-encoders and bi-encoders. Bi-encoders “encode” the item and the query independently (self-attention) and present them as vectors for downstream ANN or kNN retrieval. Whereas, cross-encoders can encode both the item and the query jointly (cross-attention), at the cost of an unfavorable inference contract at the item-query combination level, which makes it computationally expensive. Both bi-encoders and cross-encoders are tuned on some supervised signal, but since cross-encoders jointly process the query and the document, they are able to learn more intricate relationships between the query and the document in order to model the supervised signal, in comparison to bi-encoders. Bi-encoders remain the architecture of choice for most EBR, allowing for precomputation of items and query embeddings independently.

When fine-tuning bi-encoders, different labeling strategies can be employed to feed the model with relevant and irrelevant query/item pairs. As exposed in [4], training a relevance model solely on click-based signals is problematic as it is prone to reproducing the popularity and exposure bias present in the training data. However, in our case, we still want to keep the reliably positive pairs (query, item) that come from a training set labeled with CTR, while knowing that the negative labels from this dataset are not reliable indicators of irrelevance [28]. There are several ways of generating negative labels from only positive data namely ranging from trivial solutions like in-batch random negative sampling (IRNS) [18] to more complicated techniques like ANCE [58], NGame [5], GISTEmbed [45] which introduce additional complexity while still reaching a ceiling of performance in cases where reliable negative examples can be procured.

Since click data also suffers from middleman bias, i.e. a form of sample selection bias [53] due to the auction procedure and contract between eBay Search and eBay Advertising, [11] explores training on Search relevance signals to train a relevance filter. This signal gives validation on our keyphrases recommendations rather than buyer queries (which are only matched to items if Search deems them to be relevant); this has been shown to be a signal superior to clicks for our use case [11]. On the same note, since our advertisers also perform a manual check of our keyphrase offering, aligning with human judgement is also critical to the adoption of our keyphrases. As a proxy for human judgement, using LLMs to generate relevance labels offers a less biased alternative, especially as they arrive pre-loaded with a broad spectrum of world knowledge, eliminating the need for domain-specific pre-training or fine-tuning [26] for our massive inventory of diverse items (2.3 billion items over 100,000 categories). For our encoders to learn from the diversity of the above-mentioned labels, multi-task training can

be employed and has seen much success in this domain. Piccolo2 [17] employed InfoNCELoss [52], CoSENTLoss, and a variation on InfoNCELoss without in-batch negatives and only hard negatives selected using BM25 [42] to reach state-of-the-art performance on Chinese.

Recent investigations, such as [13, 14, 32, 50, 54], examine the role of Large Language Models (LLMs) in generating labels to enrich search data and enhance retrieval. This method underscores the scalability of LLM-produced labels in minimizing the high costs of manual annotation. Beginning with a limited human-annotated dataset, researchers refined an LLM to produce an amplified set of labels to train a more compact model. LLMs, while gaining traction as assessment and data augmentation instruments, encounter critique about their suitability for evaluation and data generation, as highlighted in [2, 44, 47]. Recent findings in [10] reveal that labels generated by general-purpose LLMs optimally can be used to fine-tune cross-encoder models for assessing keyphrase relevance better than those derived from search logs or fine-tuned LLMs. It strongly advocates for the adoption of business-oriented metrics for relevance model evaluation, providing more actionable insights into model efficiency and business outcomes.

To effectively distill knowledge from LLM signals, training on soft outputs would be beneficial compared to training on the hard outputs from the LLM. Cross encoders are known to learn binary labels with much better accuracy. TwinBERT [30] and PROD [29] introduce the idea of distilling cross-encoders into a twin tower BERT model. TwinBERT processes input embeddings independently and then applies a crossing layer to determine the final score, using either a residual network or simple cosine similarity. PROD outlines a framework for progressive teacher and data distillation, where both model and data complexity are incrementally reduced. Similarly, ERNIE-search [31] adopts a method that uses a Teacher-Assistant [33] framework to distill from a cross-encoder (CE) to a late-interaction model like ColBERT [23, 43], and eventually to a bi-encoder (BE). Following ERNIE-Search, distilling knowledge from a LLM teacher to bi-encoder student using a cross-encoder assistant is a natural follow-up.

Although data augmentation/knowledge distillation from a cross-encoder to bi-encoder is nothing new [49] it has faced shortcomings — CUPID [3] states that the traditional pointwise MSE loss [24] for distillation does not work on cross-encoder outputs for distillation to bi-encoder. D2LLM [26] explores distilling knowledge from a LLM cross encoder into a bi-encoder (augmented with an Interaction Emulation Module) by using a multi-task training scheme that includes a Pearson-based rank imitation loss — which counters the claim made by CUPID [3] as to the distillation of cross-encoders to bi-encoders using simple pointwise losses. In summary, our paper explores navigating the various biases presented in click-data in the context of advertiser keyphrase recommendations and explores training on disparate signals from Search relevance scores and LLM labels. This framework of multi-task learning also is supplemented by a Teacher-Assistant framework using an LLM teacher, cross-encoder assistant and a bi-encoder student. Our paper also covers a multitude of ablation studies exploring the various loss function and labels in the multi-task framework and the various loss functions in the context of knowledge distillation from cross-encoders to bi-encoders.

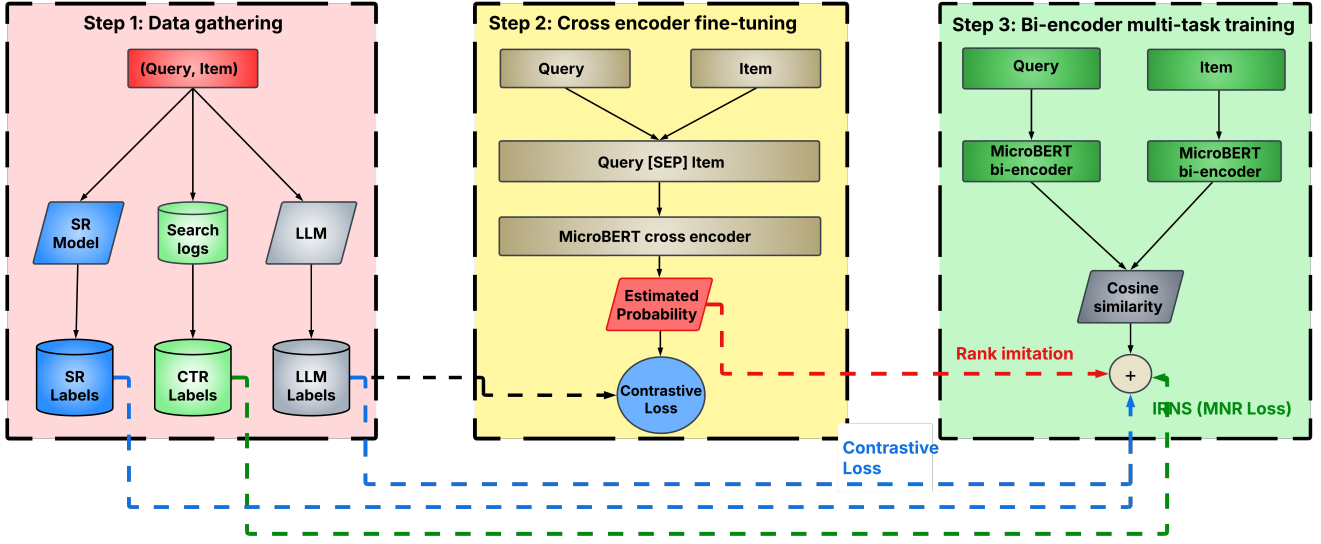


Figure 3: Our proposed architecture for multi-task knowledge distillation. The LLM is distilled to a cross-encoder, which is in turn distilled to the bi-encoder via multi-task hybrid training

3 Embedding Based Retrieval

A dual-tower architecture that independently processes keyphrases and items serves as an excellent framework for a recall model, especially when the focus is on delivering budget-conscious recommendations with acceptable latency. To mitigate biases present in click-based ground truth data, we supplement our dataset with additional signals derived from Large Language Models (LLMs) and search relevance scores, employing a hybrid training approach as illustrated in Figure 3. Within this framework, a cross-encoder is utilized to refine insights from the LLM-generated labels. Our discourse will commence with an outline of our cross-encoder architecture before delving into the bi-encoder design. Following the detailed discussion of these components, we shall elaborate on the multi-task training procedure that we have previously mentioned. Prior to engaging with these sections, we will provide a comprehensive overview of the methods employed in compiling our keyphrase-item dataset, establishing the foundation for the subsequent analysis.

3.1 Dataset curation

In this study, we systematically collected a range of data points encompassing user-query interaction records (referred to as click-data), Search Relevance (SR) metrics, and relevance scores derived from Large Language Models (LLM), all based on item-keyphrase pairs. Notably, click-data arises from item-queries (keyphrases) that successfully pass through a search relevance filter, commonly influenced by intermediary biases, and is inherently subject to biases introduced by the ranking mechanisms on the search side, known as sample selection bias. Conversely, both the SR and LLM datasets are item-keyphrase pairs that are generated through our recommendations, thereby inherently free from any form of sample selection or middleman biases.

3.1.1 CTR-based labels. As mentioned above, we collected the CTR scores for a (query, item) pair as the ratio of the total number of clicks over the total number of impressions on search activity logs from the past 30 days. Given a CTR ratio, the corresponding (query, item) pair was labeled as positive if the ratio is above the threshold of 0.05. As explained in [34], a low query-item CTR does not necessarily mean that the item is irrelevant for the query. Therefore, when using the CTR-based labels, *the positive labels are reliable but the negatives are not*. The minimum thresholds of CTR, impressions and clicks are used to eliminate noise such as 1 click 1 impression or 1 click and 1000 impression. An important thing to note is that due to the nature of the auction process, all click data is deemed relevant by eBay Search (i.e. they are only surfaced to buyers because it is deemed relevant by eBay Search). The dataset size for the click-based labels is 10,702,747 records.

3.1.2 Search Relevance labels. We also gathered relevance scores from eBay Search annotated during the auction process for (item, keyphrase) pairs from our keyphrase offering. These scores are the output of our Search Relevance (SR) model which takes buyer context into account, the scores reflect an average score distribution within a time period. Using these scores, we label a (query,item) pair as positive if its SR score is above a certain threshold (determined by business metrics for our promoted placement and varies across different countries), and negative otherwise. The training dataset comprises of 18,721,682 records.

3.1.3 LLM labels. We generated relevance judgements using Mixtral 8X7B [19] for each (item, keyphrase) pair.² Mixtral 8X7B demonstrates a 90% concordance with click data, which serves as an indicator of positive human judgments, with a fair level of agreement

²Other “open-source” models such as LLAMA 2 [51], DBRX [41], and Qwen-2 [59] were considered but faced distillation and licensing constraints for commercial use. For commercial models like GPT-4 [37, 38] usage is often hindered by rate limits or API call restrictions to external sources.

with independently collected human judgment data — see [10]. The training set is identical for the SR labels while the test set is 3,524,414 records. The prompt is illustrated below:

Prompt Design

Below is an instruction that describes a task. Write a response that appropriately completes the request.

Instruction:

Given an item with title: "{title}", determine whether the keyphrase: "{keyphrase}", is relevant for cpc targeting or not by giving ONLY yes or no answer:

Response:

3.2 The cross-encoder

We first formulate the use of the cross-encoder in the context of keyphrase recommendations, and then introduce details about our training setup and our choice of architecture.

3.2.1 Problem formulation. The cross-encoder is a transformer model that jointly handles pairs of input sentences by applying cross-attention mechanisms. This is achieved by concatenating the sentence pair into a single sequence, which is subsequently fed into the transformer architecture. The model then produces an integrated representation for this concatenated sequence. Following this, a classification layer makes the final prediction regarding the label. For the cross-encoder inputs, one input is the user-provided keyphrase, while the second consists of both the item title and its corresponding category, combined together. Consequently, the comprehensive input delivered to the cross-encoder is structured as query [SEP] category name [SEP] item title.

3.2.2 Training. The base model we used for the cross-encoder is the microBERT model, a distilled version of eBERT [6] (pre-trained on a dataset that includes eBay item title corpus) with the architecture of mobileBERT [48]. It is a compact and efficient version of eBERT that retains high accuracy while significantly reducing model size and inference latency. More precisely, it is 4.3× smaller and 5.5× faster than eBERT while achieving comparable performance. We fine-tuned the cross-encoder on the labels coming from the LLM model of 50,078,315 records, with the cross-entropy loss on the dataset described above. When evaluated on a test set of 7,503,031 (item, keyphrase) pairs, it yielded a F1 score of 91% with a precision of 92% and recall of 90%. The cross-encoder achieved a F1 score of 96%, thus validating its use as an assistant model.

3.2.3 Inference phase. After the training phase described above, we used the fine-tuned cross-encoder to give scores for each sentence pair in the bi-encoder training set.

3.3 The bi-encoder

The bi-encoder model is a traditional BERT-based [9] bi-encoder model which encodes items titles (concatenated with their meta category name) on the one hand, and buyers keyphrases on the other. The projections from the items and the keyphrases are then compared using *cosine similarity*. Within this architecture, any

transformer model can be used as the base model. Each input is first passed through the base transformer model, which produces token-wise contextualized representations. After this step, a mean pooling operation is applied to condense the token representations into a fixed-length vector. The choice of the base model significantly influences the quality of the derived embeddings. In the next section, we present the different base models that we have tried as part of our experiments.

3.3.1 The base model. We have experimented with 3 models, eBERT, MicroBERT and ModernBERT.

eBERT: Multilingual eBERT model [6], pre-trained on eBay item data and general domain. The architecture used is a BERT-base configuration with 12 layers, outputting an embedding of dimension 768.

MicroBERT: Compressed and distilled version of eBERT (around 4 times smaller, and around 5 times faster; trained with a procedure explained here [48]). It achieves a smaller size due to a smaller intermediate layer (size of the feedforward layer inside the transformer), of 384 compared to 3072 for the original; Output embedding dimension is still 768.

ModernBERT: [56] We used a version of modernBERT that is made multilingual through trans-tokenization and cross-lingual vocabulary transfers [40]. However, this base model was *not pre-trained on eBay data*. ModernBERT features many improvements over the original BERT architecture, including: longer sequence length (8192 tokens, compared to 512 for original BERT), the use of Rotary positional embeddings instead of absolute ones, alternating global and local attention (every third layer uses global attention; the rest use local sliding window attention - all of these use Flash attention). Generally speaking, it is deemed a better model than the original BERT model, with an overall GLUE score of 88.5 (compared to 80.5 for BERT-base).

3.3.2 The training process. As explained above, we would like to fine-tune a student bi-encoder model with a training scheme that *simultaneously* includes a rank imitation loss based on the assistant cross-encoder output and a multi-task hybrid training scheme based on separate ground truth labels. We therefore used different loss functions for each of our labels and put them together in a multi-task training process where each training/evaluation batch only contain samples from one of the datasets. For a training and evaluation batch, each dataset was sampled in proportion to its size. In the section below, we review the different labels and loss functions that we used for our multi-task training scheme.

Multiple Negatives Ranking Loss. The Multiple Negatives Ranking (MNR) Loss [16] is well-suited to cases where only positive pairs are available, as it does not require manually labeled negative samples. When fed with item-keyword pairs of positive examples, this loss uses one item as its anchor, uses its given keyword as a positive example, and considers all other keywords in the training batch as negative for this anchor item (IRNS). This approximation works well with highly-sparse datasets such as e-commerce and web datasets.³ In our use-case, as explained in [34], CTR-based

³While a better solution would be to employ better hard negative mining as illustrated in D2LLM, ANCE and other works; due to the size of our dataset the process proved

signals provide reliable positive sequence pairs, but not reliable negative pairs. Therefore, we used the MNR loss on the CTR-based labels.

$$\mathcal{L}_{\text{MNR}} = -\log \frac{\exp\left(\frac{\mathbf{z}_i \cdot \mathbf{z}_j}{\tau}\right)}{\sum_{k=1}^K \exp\left(\frac{\mathbf{z}_i \cdot \mathbf{z}_k}{\tau}\right)} \quad (1)$$

where:

- $\mathbf{z}_i$ and $\mathbf{z}_j$ are the embeddings of the positive pair,
- $\mathbf{z}_k$ is the embedding of a negative sample,
- τ is the temperature parameter,
- K is the total number of negative samples.

Contrastive Loss. Contrastive loss [15] explicitly optimizes the embedding space by bringing similar sentence pairs closer together and pushing dissimilar pairs apart. This loss function is therefore well-suited to cases like ours, that rely on Approximate Nearest Neighbor search at prediction time. We used this loss function on both our LLM labels and our SR labels (which both include positive and negative examples). Mathematically this loss is defined as:

$$\mathcal{L}_{\text{Contrastive}} = \frac{1}{2} \left(y \cdot d(u, v)^2 + (1 - y) \cdot \max(0, m - d(u, v))^2 \right) \quad (2)$$

where:

- y is a binary label: $y = 1$ if the pair is similar, and $y = 0$ if the pair is dissimilar.
- $d(u, v)$ is a distance function (cosine distance in our case).
- m is a margin hyperparameter that sets the minimum required separation for dissimilar pairs.

This loss function encourages smaller distances for similar sentence pairs ($y=1$) and larger distances for dissimilar pairs ($y=0$).

Pearson correlation Loss. As shown in [26], maximizing the Pearson correlation between the student’s logits and the teacher’s logits enables the student model to replicate the teacher’s subtle ranking nuances. It does that by minimizing the Pearson rank imitation loss, defined as:

$$\mathcal{L}_{\text{RI}}^{\text{Pearson}} = 1 - \text{corr}(z_T, z_S) \quad (3)$$

where z_T and z_S are the scores from the teacher and the student.

In [26], this loss is especially employed on BM25 hard negatives. In our case, we already have the hard negatives from the teacher LLM and subsequently from the cross-encoder assistant and don’t require any hard negative filtering. We therefore used this loss directly on all of the student’s logits and teacher’s logits.

CoSENT Loss (Cosine Sentence Loss). This is another loss used during knowledge distillation [22]. Mathematically, it is computed as:

$$\mathcal{L}_{\text{CoSENT}} = \log \sum_{(i,j),(k,l)} (1 + \exp(s(i, j) - s(k, l)))$$

to be quite expensive. We acknowledge a potential for improvement in this area and leave this up to future research.

Here, (i, j) and (k, l) are any input pairs in the batch such that the cross-encoder-based similarity of (i, j) is greater than (k, l) . s is the bi-encoder-based similarity function.

MSE Loss. This is the traditional MSE loss, calculated as the Mean Squared Error between the cross-encoder similarity scores, and the cosine similarity scores for the bi-encoder embeddings for items and keyphrases.

$$\mathcal{L}_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^N (y_i - \cos(u, v)_i)^2 \quad (4)$$

where u and v are the embeddings for the item and keyphrase respectively and y is the score of the cross-encoder.

3.3.3 Matryoshka embeddings. Embedding dimensionality directly impacts latency and computational cost. Using Matryoshka embeddings [25] significantly reduces the cost of nearest-neighbor search while not significantly impacting retrieval accuracy and precision. Matryoshka Representation Learning (MRL) is a technique that enables learning representations of varying sizes within a single high-dimensional vector. Instead of training multiple models for different embedding sizes, MRL optimizes a nested structure where the first m dimensions of an embedding are designed to be as informative as an independently trained m -dimensional representation. This hierarchical structure is achieved with minimal computational overhead and ensures efficient deployment across different tasks [25]. This is achieved by modifying the loss function: in addition to the initial loss, the loss value for each Matryoshka dimensionality is added to the overall loss function. We reduced our dimension size to 64 for faster ANN.

4 Experimentation and Ablation Studies

In this section, we present our offline experiments and ablation studies. Please note that, due to time and resource constraints, we performed our ablation studies on a subset of the 3 million (item, keyphrase) pairs test set. The few evaluations that we did run on the entire dataset were directionally similar to the ones observed on the smaller test set, with a F1-score improvement from 67% to 83% (between the baseline model and our proposed solution). Furthermore, as many different losses and architectures need to be compared, we chose to evaluate all of them on the *LLM-labeled test set only*, for reasons and clarity and readability. At the end of this section, we present our offline evaluation algorithm, which mimics production settings and incorporates estimations of uniqueness and diversity, similar to the detailed evaluation scheme presented in [34].

4.1 Modifying the base transformer model

In this first analysis, we have fine-tuned our bi-encoder model using the LLM-labeled training set with the contrastive loss function. The only parameter that we have changed here is the base model. We report the classification metrics we observed when using microBERT, modernBERT and eBERT as our base model.

As shown in Table 1a, both the microBERT and the eBERT models give better performance than modernBERT, even though modernBERT has a higher GLUE score than BERT (88.5 vs 80.5) and a

Base models	Recall	Precision	F1
MicroBERT	0.92	0.78	0.85
eBERT	0.92	0.81	0.86
ModernBERT	0.91	0.76	0.83

(a) Changing the base model

K.D. Loss	C.E. corr	F1	Precision	Recall
MSE	0.62	0.82	0.75	0.91
CoSENT	0.76	0.88	0.86	0.90
Pearson Loss	0.79	0.89	0.87	0.91

(b) Changing the KD Loss

Table 1: Ablation studies: (a) base models and (b) KD loss

much higher context length (8192 vs 512). This result illustrates the importance of pre-training, as the modernBERT version that we used here was not pre-trained on eBay’s vocabulary.

Table 1a also shows that using eBERT as our base encoder yields slightly better results than microBERT. This is expected, as microBERT is a distilled version of eBERT. Due to the size of our dataset, we chose to use microBERT for the rest of this study, as it shortens the batch prediction time by 30% on average.

4.2 Assessing the multi-task framework

In this section, we present the influence of multi-task training on the bi-encoder performance. In previous studies involving multi-task learning [26], we felt the effect of each component was not clearly studied — albeit there is a limited amount of ablation studies one can perform. In this study however we set out to perform a more comprehensive set of ablation studies to segregate each section of the multi task framework. We first started with the basic CTR labels and then started adding the contrastive labels (SR and LLM) and then added the KD cross-encoder scores distilled from LLM for additional signal and more accurate calibration.

We also additionally test a variety of KD losses. During the knowledge distillation process, the bi-encoder is trained on the soft outputs of the cross-encoder. Therefore, it is valid to evaluate this process based on the *Pearson correlation* between the soft outputs of the bi-encoder and those of the cross-encoder⁴. We also evaluate these losses with the bi-encoder’s F1 score on the test set. In CUPID [3] it was stated that MSE loss performs poorly for distillation from cross encoder to bi-encoder; we verify this in our own observations in Table 1b. Interestingly, we also tried other KD losses, like CoSENT which is technically a pairwise ranking loss with a calibration component to it, and the Pearson Correlation loss described in D2LLM [26] which has a batch-wise ranking calibration component to it expanding over the pairwise construction of CoSENT. Interestingly, the batch-wise rank imitation loss of Pearson-Correlation performs better than the pairwise CoSENT Loss which performs better than the pointwise MSELoss.

4.3 Offline Evaluation

The retrieval models at eBay are stacked with each retrieval model serving a different purpose (see [34] to review a comprehensive

⁴We refer to this correlation as C.E. corr in our table

Labels	median kw cnt	LLM pass rate
LLM+CTR+KD	12.0	70.57%
LLM+SR+KD	12.0	51.30%
LLM+KD	11.0	48.81%
LLM	11.0	60.85%
LLM+SR+CTR+KD	11.0	69.94%
KD	11.0	39.02%
SR+KD	11.0	45.79%
CTR	7.0	59.65%

Table 2: Ablation study - Effect of Labels and their corresponding training components. All models are retrieved with k=20.

summary of what models are in production). Hence to obtain a comprehensive estimation of our solutions incremental impact, we must exclude keywords already suggested by other retrieval models in production — albeit the EBR (CTR) model we plan on replacing. Following this de-duplication, the remaining keywords are then passed through the downstream eBay Advertising relevance filter. Upon completing these steps, we can estimate the *median count of de-duplicated relevant keywords per item* (median kw cnt in Table 2) which are surfaced to the sellers. These procedures are outlined in algorithm 1. To also gain insights on alignment on seller judgment and search judgment a sample of 10000 records per model (after passing the relevance filter) is then put through our LLM (*Mixtral-8x7B-Instruct-v0.1*) and Search Relevance Model. What we observed was that for Search judgment we got a more than 99% alignment for all the models, indicative of auction efficiency that our model would bring. From our observations recorded in Table 2 we see that the LLM+CTR+KD and LLM+SR+KD labels perform the best in terms of efficiency with a median keyword 12 after the relevance filter that was surfaced to the sellers. In addition, conferring with LLM judgment which serves as a proxy for seller judgment we see that LLM+CTR+KD has the best performance with 70.57% of its keyphrases passing the LLM judgment, meanwhile the LLM+SR+CTR+KD is a close second. In lieu of these results we decided to test the best overall model LLM+CTR+KD in online settings.

Algorithm 1 Evaluation in Production settings

- 1: **Sample** a certain number of items from different sites.
- 2: Call all the other recall models on these sampled items.
- 3: **Embed** all items and keywords with the fine-tuned bi-encoder.
- 4: **for** each item **do**
- 5: Get the top k closest keywords in the embedding vector space.
- 6: Pass the resulting k keywords through the relevance model or the LLM.
- 7: Out of the resulting keywords from the previous step, take only the ones that are **not** in any other recall lists.
- 8: **Calculate** the median of this metric over all the items from step 1.

5 Production System Design

The production architecture depicted in Figure 4 comprises two main parts: *Near Real-Time (NRT)* Inference and Batch Inference. Batch inference handles items with a delay, while NRT prioritizes immediate items, particularly those newly created or updated by sellers. Batch inference has two components: 1) full batch inference

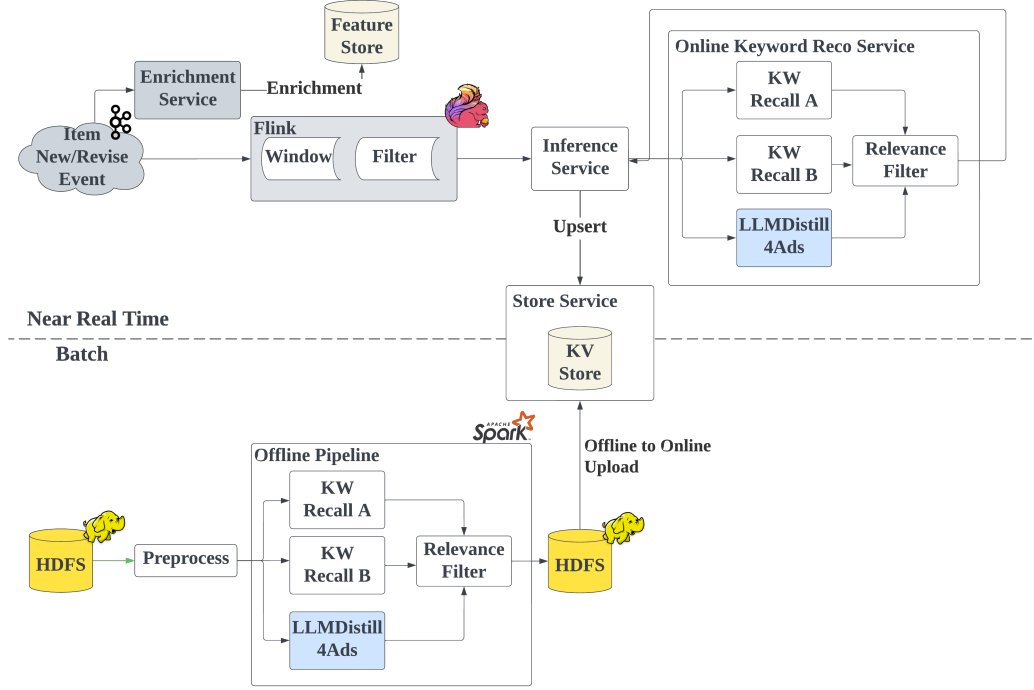


Figure 4: Production Serving Architecture for keyphrase recommendations.

for all items, and 2) daily differential (Diff) to integrate new and updated items with existing data. NRT inference utilizes triton and onnx serving using V100 GPUs, activated by item creation or updates managed by Flink processing and feature enrichment. The full batch handles approximately 2.3 billion items, while the daily Diff supports a churn of 20 million items. As the full batch runs just once, Diff latency determines model deployment viability, being about 35 minutes for bi-encoders. The ANN job downstream on a takes an additional 2.5 hours daily and for NRT our vector database service helps in that regard. Latency numbers reported for our batch inference use PySpark [61] (1500 executors, 20g memory, 4 cores), leveraging transformers [57] and onnxruntime [8].

6 Impact

We ran an A/B test with our new model as the treatment replacing the CTR only EBR model for 12 days in the US market. While we observed directional improvement in clicks (11.5%, $p = 0.25$), it was not statistically significant. However, we do see some statistically significant improvement in terms of GMB (Gross Merchandise Volume Bought, i.e. sales) which increased by 51.26% ($p = 0.01$) implying the better relevance of items converting same number of clicks to more sales. The return on advertising expenditure ($ROAS = \frac{GMB}{Ads\ Revenue}$), which marks the bottom line for the seller — the ratio of sales money to the money spent on advertising (which is proportional to clicks) experienced a notable improvement of 38.69% ($p = 0.02$).

7 Conclusion

This research thoroughly examines the constraints encountered when exclusively relying upon click-based indicators to fine-tune bi-encoder models specifically for the classification of sentence pairs within the e-commerce sector. Our investigations reveal that substituting traditional Click-Through Rate (CTR) signals with those generated by Large Language Models (LLMs) produces a marked enhancement in model performance. Furthermore, additional gains can be achieved by engaging in an intermediate cross-encoder model during the fine-tuning phase via knowledge distillation techniques. Notably, conducting knowledge distillation concurrently with training on supplementary *raw* labels confers further performance enhancements for the bi-encoder models. Our analysis further underscores the significance of the initial pre-training of the base model. Such preparatory steps enable comparatively smaller architectural models to surpass their more sizable counterparts in performance metrics. Moreover, our research identifies a rank imitation loss based on Pearson correlation as an exceptional knowledge distillation loss function, which notably outperforms both CoSENT and Mean Squared Error (MSE) loss functions. To supplement our findings, we introduce a rigorously structured evaluation protocol aimed at quantifying the business impact of potential candidate generator models under conditions that closely replicate those found in practical production environments.

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8 Appendix

8.1 KD Losses

Following on this logic of MSE (pointwise) vs CoSENT (pairwise) vs. Pearson Correlation Loss (batch-wise), we tried additional advanced ranking losses, namely neural NDCG [39] and Lambda Loss [55], however, the results were horrible (less than 0.1 in recall and precision). Our suspicions are that these losses are generally applied with a seed query and have a rank misclassification penalty for disparate ranks — which might cause some issues, as the cross-encoder was never trained for specific ranking calibration, rather overall linear directionality and calibration appear to be main drivers of performance. This direction needs further research.

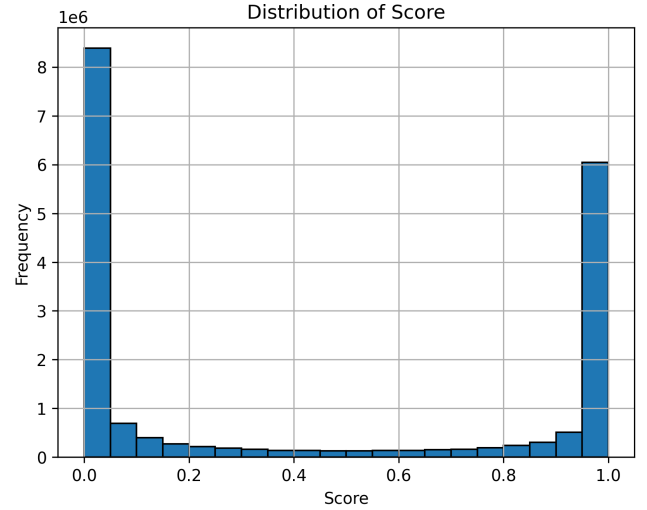
8.2 Fine-tuned vs General LLM

During LLM-as-a-judge data augmentation, we were faced with the option of using General vs fine-tuned LLM. General LLMs have an advantage in being less prone to biases that affect specialized models. Conversely, fine-tuned models are fine-tuned on human judgment and can propagate the biases present in the original small human judgment sample present. In a previous study [10] we found that the general LLM was more aligned with our business metrics than the fine-tuned LLM, although the fine-tuned LLM showed better alignment on the small amount of human judgment data. We later discovered problems with our human judgment data collection, and use general LLMs until we fix the issue with our human judgment data.

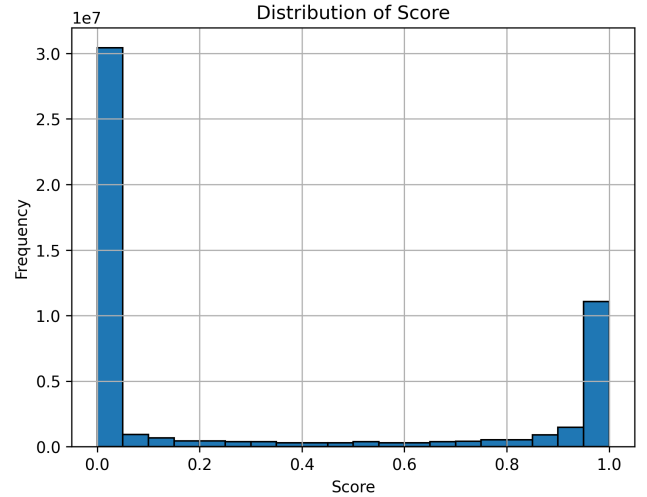
8.3 LLM Logits vs cross-encoder

In our study we could have skipped the cross-encoder and went directly for the LLM teacher generating the scores through logits of yes/no through constrained decoding. However, in our previous study [10] we found this to be not very helpful. We even tested this on our current EBR models and found results to be highly unsatisfactory. The main reason is the distribution of scores of cross-encoder which learns from the yes/no of the LLM and the softmax of the logits of the LLM as shown in Figure 5. As seen in the figure the cross-encoder has a much more even distribution for the extreme values of 0 and 1 where the golden class distribution is 50-50, whereas for the logits we see a much more disparate distribution.

This is then propagated to downstream models which worsens the learning.



(a) Cross-encoder score distribution



(b) Softmax of logits of the LLM distribution

Figure 5: Kernel Density Estimation of the scores of cross-encoders and LLM probabilities.